Tangents

1 $f'(2) = 17$

2

a $\frac{dy}{dx} = x^2 + x^5 + 2$

b $-16\,756$

3 $f'(4) = -1$

4

a $x = -5, x = 2$

b $f'(2) = 7$

5

a $\frac{dy}{dx} = 8x - 5$

b $x = \frac{5}{8}$

6

a $f'(2) = -12$

b $y = -12x + 16$

7

a $f'\left(\frac{1}{9}\right) = \frac{15}{2}$

b $y = \frac{15x}{2} + \frac{5}{6}$

8

a $f'(4) = \frac{3}{2}$

b $y = \frac{3x}{2} + 6$

9

a $f'(-2) = -\frac{3}{8}$

b $-\frac{1}{4}$

c $y = -\frac{3x}{8} - 1$

10

a 8

b $y = 4x + 4$

11

a To find the equation of the tangent line at $x = 2$, we need to find the gradient of the tangent line at that point, which is given by $f'(2)$. We also need to find the y-coordinate of the point at $x = 2$. We can then substitute the known gradient and point into the equation of a line ($y = mx + c$) to find the value of c.

b $y = 36x - 48$

12

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

a $\dfrac{5}{3}$

b $y = -\dfrac{4x}{3} + \dfrac{1}{3}$

13

a $g'(x) = 20x^4 - 12x^3 + 6x^2 - 7x^{-3}$

b $y = 7x - \dfrac{1}{2}$

14 $y = 5x - 4$

15 $y = 7x - 19$

16 $(15, 2025), (-11, -2057)$

17 $(2, -16), (-4, 62)$

18 $(9, -486)$

19 $x = \dfrac{3}{4}, x = -2$

20 $(2, 4)$

21 $(2, 8)$

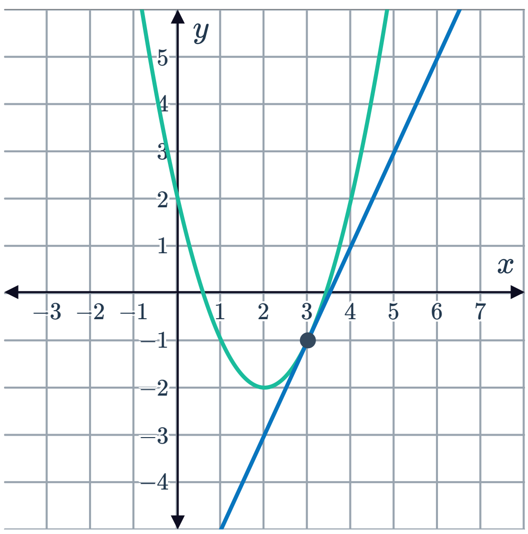
22 $y = -13$

23

a 2

b $(2, -2)$

c

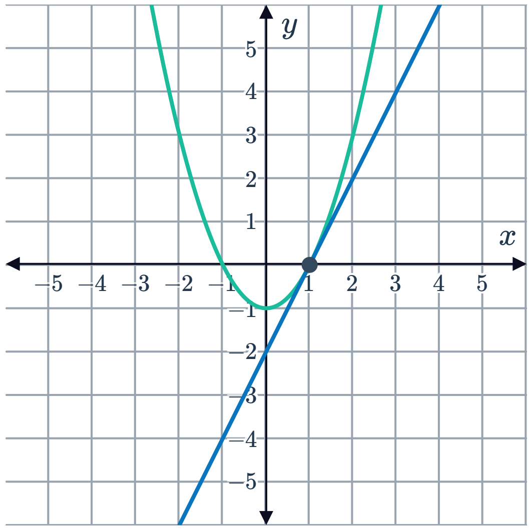


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

24

a 2

b



25

a $x=\dfrac{1}{2}, x=\dfrac{1}{6}$

b At $x=\dfrac{1}{2}$ the tangent is $y=-\dfrac{1}{4}x$ and at $x=\dfrac{1}{6}$, the tangent is $y=-\dfrac{x}{4}+\dfrac{1}{54}$.

26 $k=40$

27

a $a=900$

b $b=90$

28

a $\dfrac{dy}{dx}=2x+b$

b -5

c $b=-23$

d $c=79$

29

a $b=14-3a$
$b=11-2a$

b $a=3, b=5$

30

a $c=0, d=-28$

b $b=-2a+7$
$b=9-3a$

c $a=2, b=3$

31

a $m=8, m=4$

b $y=8x-22$

c $y=4x-10$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Normals

32

a $\frac{dy}{dx} = 50x^4 - 8x^3 + 24x^2 - 12x + 4$

b 4018

c $\frac{-1}{4018}$

33

a 2

b $-\frac{1}{2}$

34

a $f'(x) = 2x + 8$

b 16

c $y = 16x - 1$

d $-\frac{1}{16}$

e $y = -\frac{x}{16} + \frac{253}{4}$

35

a $y = 32$

b $x = 4$

36

a $\frac{dy}{dx} = 2x + b$

b 4

c $\frac{63}{4}$

d 30

Applications

37

a $f'(x) = -2x$

b -6

c $y = -6x + 45$

d $\frac{1}{6}$

e $y = \frac{1}{6}x + \frac{53}{2}$

f $x = \frac{15}{2}$

g $x = -159$

h $\frac{333}{2}$ units

i $\frac{8991}{4}$ units^2

38

a $\frac{dy}{dx} = 3x^2 - 20x + 25$

b 13

c $-\frac{1}{13}$

d $x = 84$

e 252 units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647